Go 2 · Build in SUMO

Build traffic-signal agents in SUMO, one lab at a time

Six short Python labs: a first agent, talking agents, a learning agent, eco-driving connected cars, priority for buses and ambulances, and a language-model agent behind a safety shield.

smrahmanbd.github.io/agentic-atsc-lab

7 ideas, each building on the last

1 Set-up and the Junction class

4 s lost to yellow plus all-red at every stage change

2 Lab 1: your first agent

20 s agent's longest wait in Lab 1 (max-pressure: 301 s)

3 Lab 2: many agents on a corridor

−7% stops per vehicle when agents talk (0.95 → 0.88)

4 Lab 3: a learning agent (Q-learning)

31 distinct states the learner visited in 30 episodes

5 Lab 4: eco-driving with SPaT and GLOSA

−55% stops per vehicle, fixed plan, 0 → 100% CAVs

6 Lab 5: tools for priority

−77% bus delay with the priority tool vs actuated (33.4 → 7.8 s)

7 Lab 6: an LLM agent behind a safety shield

165 bad answers from the mock model; no unsafe one reached the signal

Three surprises to start

The cleverest-sounding controller left a car waiting 301 s.

In Lab 1 max-pressure, a well-known efficient rule, ignored how long the side street had waited. One car waited 301 s before SUMO removed it. The agent, with a small weight on waiting time and a 90 s cap, never let anyone wait more than 20 s.

Several results here are honest losses.

Talking agents did not beat actuated control on delay; the learning agent did not beat the rules; a promise made for connected cars cost the signal delay. Knowing where a method does not win is part of learning it.

The language model is a mock, on purpose.

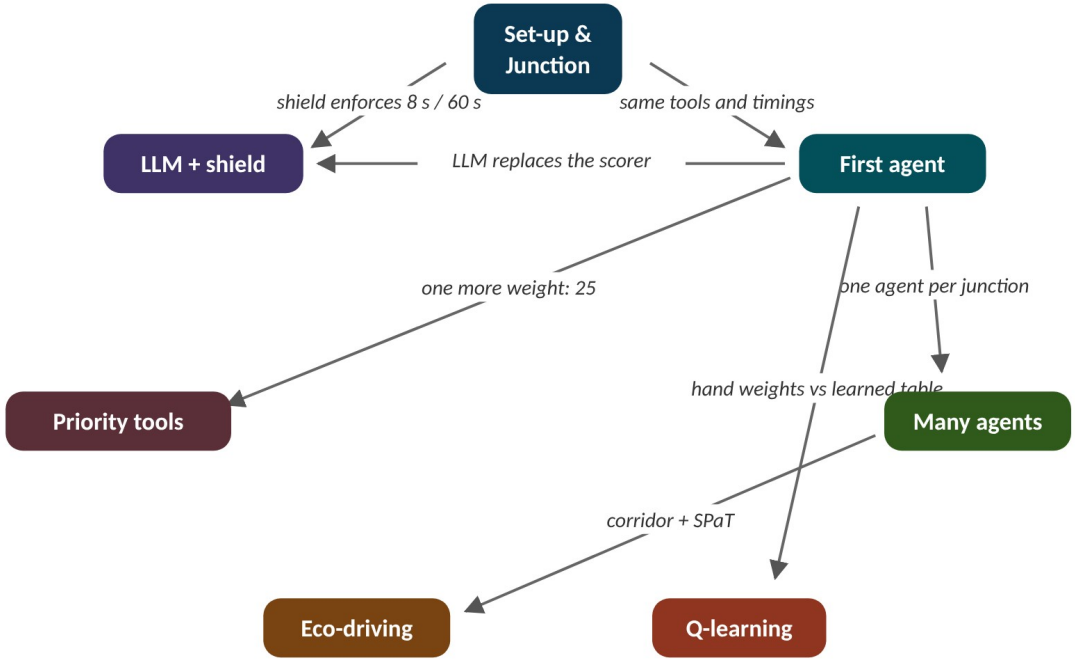
Lab 6 runs offline with a pretend model that gives bad answers 20% of the time. That makes the lab repeatable and lets you watch the safety shield do its job. A real model can be plugged in with one flag.

Words you will meet

SUMO	A free traffic simulator that moves every vehicle second by second.
TraCI	SUMO's remote control: Python reads the traffic and sets the lights through it.
Stage	A set of movements green together. Here: EW (main road) and NS (side street).
Agent	A controller that perceives, reasons and acts toward goals, and explains itself.
Observation	What the agent sees at one moment: queues, waits, arrivals, messages.

Delay	Time a trip lost compared with driving freely, averaged per vehicle.
Platoon	A bunch of cars travelling together, usually released by the previous green.
CAV	Connected (and automated) vehicle: a car that can receive signal timing and advice.
LLM	Large language model: software that reads and writes text, such as a chatbot.

The map of today's ideas



1 Set-up and the Junction class

IN PLAIN WORDS

Think of a cooking contest where every chef gets the same oven, the same knives and the same timer. Whatever dish wins, you know it won on the recipe, not because one chef had a better stove.

- SUMO simulates, TraCI is the remote control
- One Junction class for every controller
- Yellow 3 s, all-red 1 s
- Min green 8 s, max green 60 s
- Seven shared perception tools

4 s

lost to yellow plus all-red at every stage change

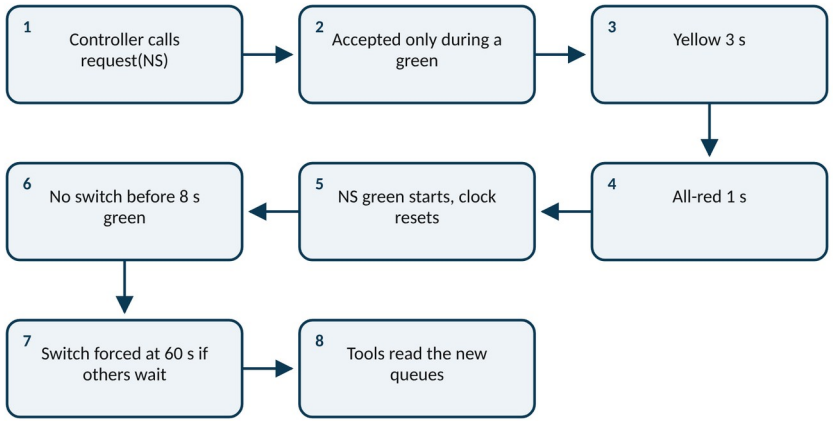
Worked example: What do stage changes cost?

lost time per stage change = yellow 3 s + all-red 1 s = 4 s
shortest possible cycle = 2 × (min green 8 s + 4 s) = 24 s

Lab 1's agent made 234 stage changes in its run.

1. Lost time per change = 3 s yellow + 1 s all-red = **4 s**.
2. Total lost = 234 × 4 = **936 s**.
3. As a share of one hour: 936 ÷ 3600 = **0.26**, about a quarter.
4. Shortest cycle: 2 × (8 + 4) = **24 s**.
5. Longest cycle (both at max green): 2 × (60 + 4) = **128 s**.

About a quarter of the hour went on changeovers, so switch only when it clearly pays.



Remember it: Set-up & Junction



PICTURE THIS · STOP 1: TOOLBOX SHOP

A shop sells one toolbox to every robot: inside are a queue ruler, a waiting stopwatch, an arrivals radar, an exit gauge, a VIP detector and blue-car goggles. The shop sign says "pip install".

TRAP

The safety timings are not part of any controller's cleverness: nobody can win by skipping yellow.

2 Lab 1: your first agent

IN PLAIN WORDS

Think of a parent with two children asking for help. They weigh who has been waiting longest, who has the most to do and who is about to arrive, and only stop helping one child when the other clearly needs it more.

- Perceive, reason, act every 2 s
- Score = weighted queue, arrivals, wait
- Switch only if ahead by 2
- Every decision logs a reason
- Agent 9.6 s, actuated 9.8 s

20 s

agent's longest wait in Lab 1 (max-pressure: 301 s)

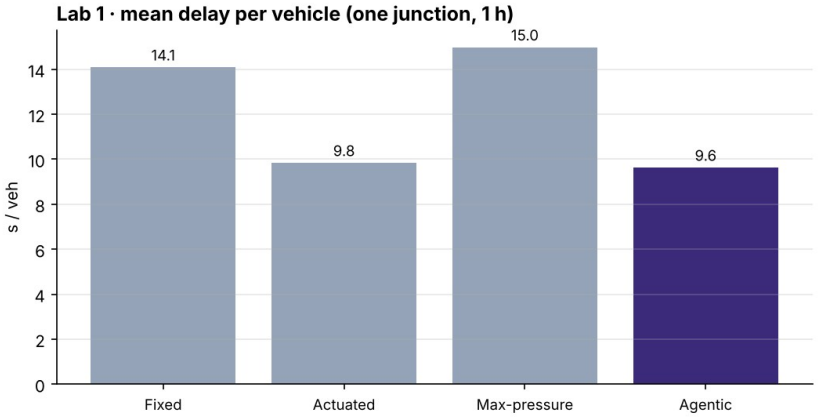
Worked example: One decision by the agent

$$\text{score(stage)} = 1 \times \text{queue} + 0.7 \times \text{arriving} + 0.15 \times \text{longest wait}$$
$$\text{switch if score(other)} > \text{score(current)} + 2$$

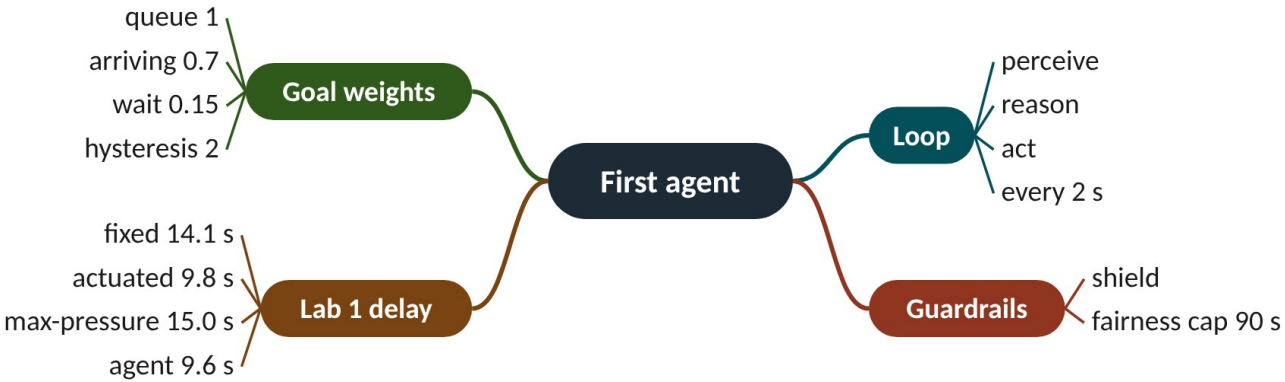
EW green 14 s. EW: 3 queuing, 2 arriving, wait 4 s. NS: 7 queuing, 1 arriving, wait 24 s.

1. EW score = 1 × 3 + 0.7 × 2 + 0.15 × 4 = 3 + 1.4 + 0.6 = **5.0**.
2. NS score = 1 × 7 + 0.7 × 1 + 0.15 × 24 = 7 + 0.7 + 3.6 = **11.3**.
3. Bar to beat = current + hysteresis = 5.0 + 2 = **7.0**.
4. 11.3 > 7.0, so the agent proposes **switch**.
5. Shield: green has run 14 s ≥ 8 s minimum, so the switch is allowed.

Log: "switch to NS: score 11.3 vs 5.0 (queue 7, longest wait 24 s)".



Remember it: First agent



PICTURE THIS · STOP 2: FIRST JUNCTION

The robot weighs two baskets on a scale every 2 seconds: cars waiting, cars arriving and how long the oldest has waited. A sticky note on the scale reads "9.6".

TRAP

Lowest mean delay is not the whole story: max-pressure's 15.0 s hid a 301 s wait.

3 Lab 2: many agents on a corridor

IN PLAIN WORDS

Think of a relay of neighbours along a street. The first sees a group of children leave school and texts the next house: "six kids, there in twenty seconds". The next one opens the gate just in time instead of guessing.

- Neighbours share platoon size and ETA
- +0.8 per car due within 12 s
- Talking: stops 0.95 to 0.88
- Actuated still lowest delay here
- Short corridor limits coordination gains

−7%

stops per vehicle when agents talk
(0.95 → 0.88)

Worked example: A platoon message, start to finish

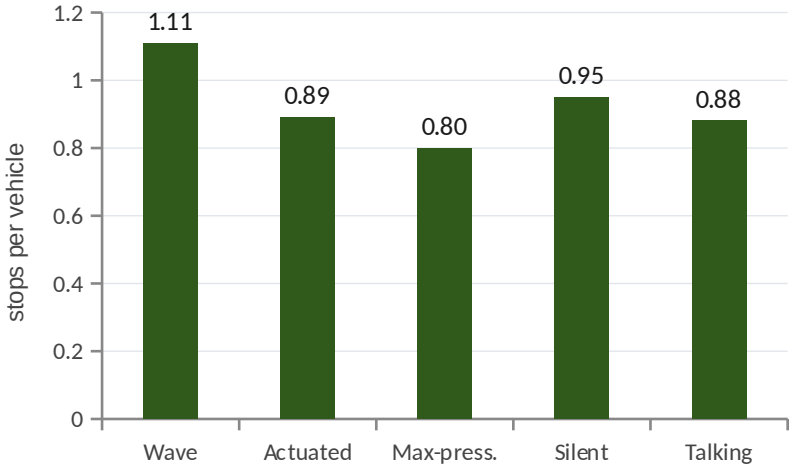
ETA = now + remaining distance ÷ 12 m/s
EW score += 0.8 × platoon size, if ETA ≤ now + 12 s

J1 sees 6 cars 240 m from J2's stop line; travel speed 12 m/s.

1. Travel time = 240 ÷ 12 = **20 s**, so ETA = now + 20 s.
2. 20 s > 12 s window, so J2 stores the message but adds nothing yet.
3. 8 s later the platoon is 12 s away: inside the window.
4. Bonus to J2's EW score = 0.8 × 6 = **4.8** points.
5. Lab 2 stops: (0.95 − 0.88) ÷ 0.95 = **0.074**, about 7% fewer.

A 4.8-point boost holds EW green for the platoon unless NS clearly leads.

Lab 2: stops per vehicle on the corridor (mean of 3 seeds).



Remember it: Many agents



PICTURE THIS · STOP 3: FOUR HOUSES

Four houses in a row pass paper aeroplanes downstream with a number and a time written on each wing.

TRAP

Talking agents did not win on everything: actuated still had the lowest delay (17.5 vs 18.2 s).

4 Lab 3: a learning agent (Q-learning)

IN PLAIN WORDS

Think of training a puppy. It tries sitting or jumping; a treat follows good choices. Over many sessions it learns which choice works in which situation. Early on it tries random things on purpose, to discover what earns treats.

- State: stage, queue bins, green time
- Reward: minus the waiting cars
- Alpha 0.2, gamma 0.9
- 31 states learned in 30 episodes
- 11.7 s: not yet beating rules

31

distinct states the learner visited in 30 episodes

Worked example: One Q-learning update

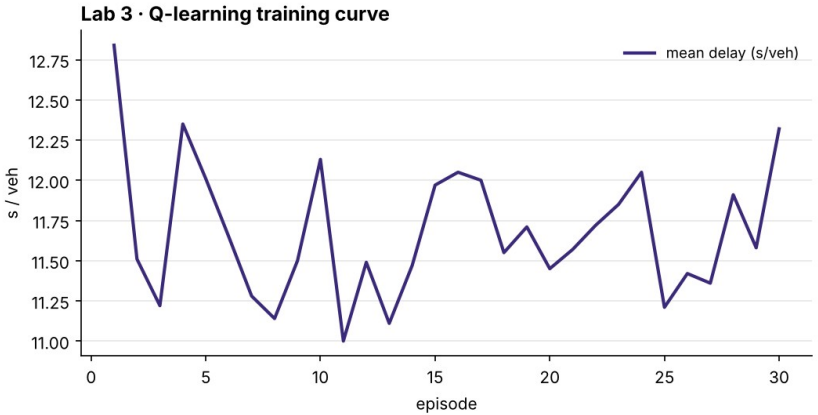
$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\, r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)\,]$$

$$r = -(\text{queue EW} + \text{queue NS})$$

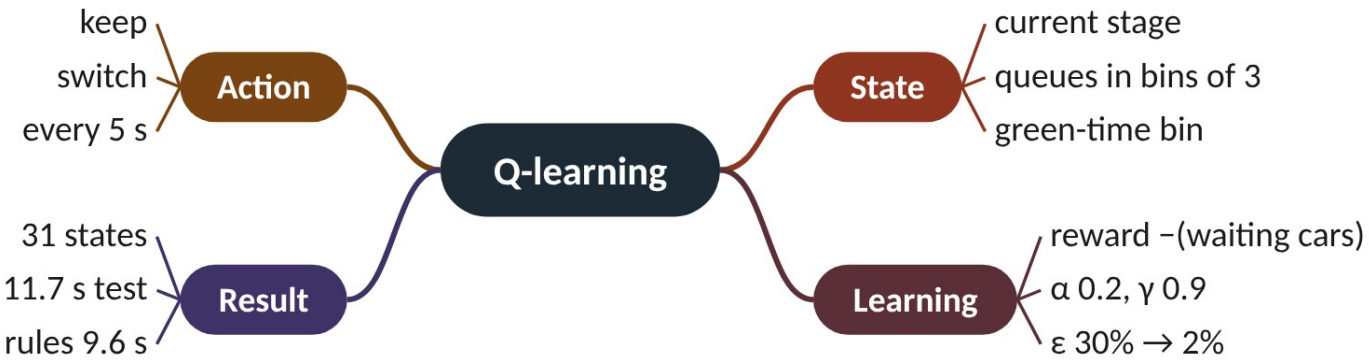
Q(s, keep) = −10. Next, 6 cars wait; best Q in the new state is −12.

1. Reward r = −6.
2. Target = r + γ × best next = −6 + 0.9 × (−12) = −6 − 10.8 = **−16.8**.
3. Error = target − old = −16.8 − (−10) = **−6.8**.
4. Step = α × error = 0.2 × (−6.8) = **−1.36**.
5. New Q(s, keep) = −10 − 1.36 = **−11.36**.

Keep now looks a little worse; only 20% of each surprise is absorbed.



Remember it: Q-learning



PICTURE THIS · STOP 4: DOG SCHOOL

A puppy learns to press "keep" or "switch" and gets a treat each time the queue shrinks; its notebook has 31 pages.

TRAP

"It learns" does not mean "it beats the rules": 11.7 s vs 9.6 s (rules) and 9.8 s (actuated).

5 Lab 4: eco-driving with SPaT and GLOSA

IN PLAIN WORDS

Think of a ferry that announces "leaving in 5 minutes". You walk calmly instead of sprinting and then waiting. If the ferry never announced its time, you would have to rush and hope.

- SPaT: stage and seconds to change
- GLOSA: distance divided by time
- Adaptive agent must commit 8 s
- Fixed plan stops: 1.12 to 0.50
- Commitment costs adaptivity

−55%

stops per vehicle, fixed plan, 0 → 100% CAVs

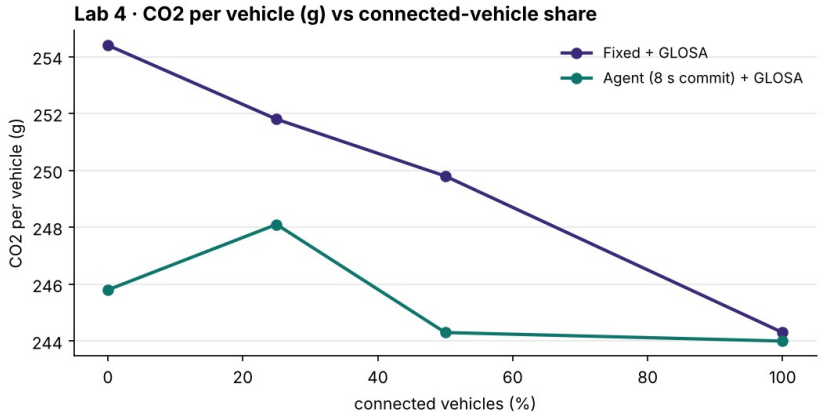
Worked example: A GLOSA speed advice

time to next green = remaining + yellow 3 s + all-red 1 s
advised speed = max(5 m/s, min(limit, distance ÷ time to next green))

CAV 150 m away, limit 13.89 m/s; NS green, committed to change in 10 s.

1. At the limit it would arrive in 150 ÷ 13.89 = **10.8 s**.
2. EW turns green in 10 + 3 + 1 = **14 s**.
3. 10.8 s < 14 s, so at full speed it would reach a red and stop.
4. Advised speed = 150 ÷ 14 = **10.7 m/s** (about 38.6 km/h).
5. Check: between 5 and 13.89 m/s, so the advice stands.

It eases to about 39 km/h and glides in on green. With no committed time, no advice.



Remember it: Eco-driving



PICTURE THIS · STOP 5: SWAN PARADE

The traffic light holds up a promise card "change in 8 s" and a parade of blue swans slows to glide through.

TRAP

SPaT is not free for an adaptive signal: to announce a time it must commit, and then cannot react.

6 Lab 5: tools for priority

IN PLAIN WORDS

Think of a supermarket that opens a till the moment a customer with a priority card appears. Others barely notice, because it happens rarely, but the card holder walks straight through.

- New tool: priority_requests() within 200 m
- Weight 25 outweighs normal queues
- Bus delay 33.4 to 7.8 s
- Ambulance 17.3 to 3.0 s
- Everyone else pays about 0.2 s

−77%

bus delay with the priority tool vs actuated (33.4 → 7.8 s)

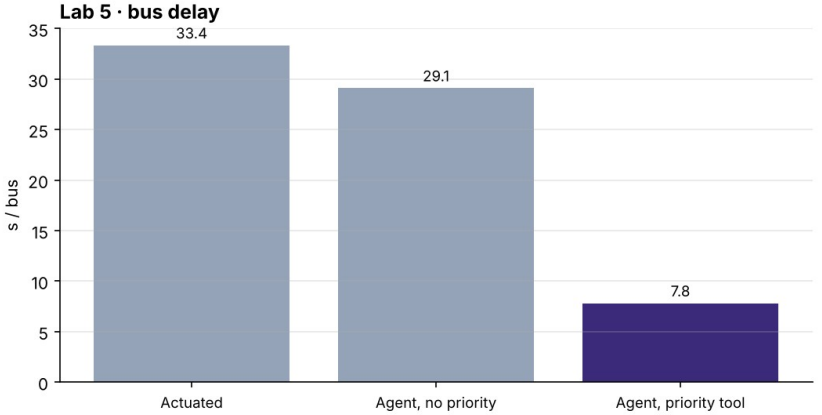
Worked example: The VIP pass beats the queue

score(stage) += 25 if a bus or ambulance is within 200 m
relative reduction = (before − after) ÷ before

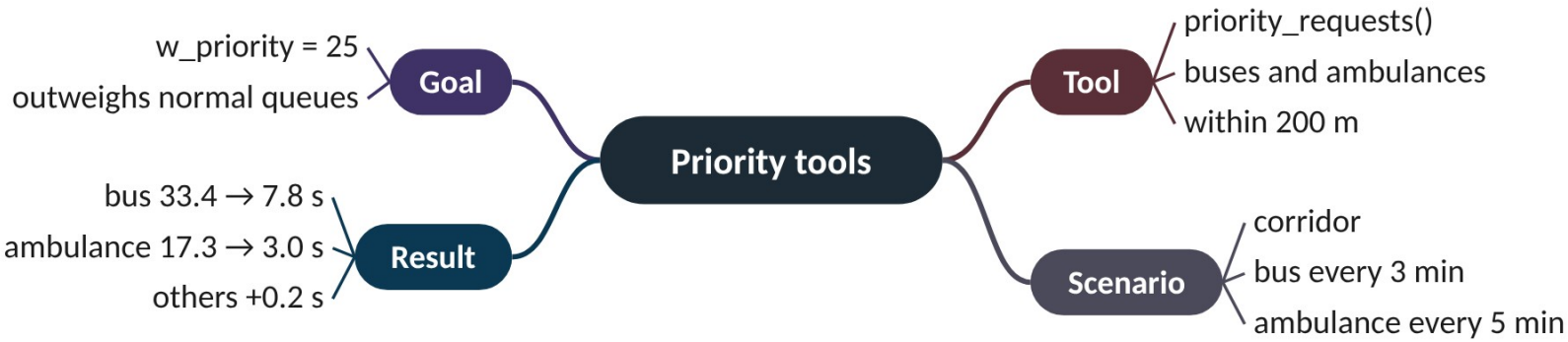
NS green 20 s: 12 queuing, 3 arriving. EW: 2 queuing, wait 10 s. A bus appears on EW.

1. NS = 12 + 0.7 × 3 = **14.1**; EW = 2 + 0.15 × 10 = **3.5**: keep.
2. EW with the bus = 3.5 + 25 = **28.5**.
3. 28.5 > 14.1 + 2 = 16.1, so **switch** (20 s ≥ 8 s, shield allows).
4. Bus delay vs actuated: (33.35 − 7.82) ÷ 33.35 = **0.77**.

One weight turns keep into switch. Vs the agent without priority the cut is (29.11 − 7.82) ÷ 29.11 = 0.73.



Remember it: Priority tools



PICTURE THIS · STOP 6: HOSPITAL ROAD

A yellow bus and an ambulance flash a golden VIP pass worth 25 points; the lights part like a red carpet.

TRAP

Priority is not free: everyone else +0.2 s, and the shield still blocks a switch before 8 s green.

7 Lab 6: an LLM agent behind a safety shield

IN PLAIN WORDS

Think of a talkative parrot giving orders on a ship, with a calm officer who checks each order against the rulebook. Sensible orders pass; dangerous ones are replaced with "hold course".

- Observation, prompt, JSON, shield, signal
- Mock model by default, repeatable
- Shield checks every single answer
- 20% bad answers: 11.6 s delay
- No unsafe order reached the signal

165

bad answers from the mock model; no unsafe one reached the signal

Worked example: What did the shield cost and catch?

observation → prompt → JSON answer → shield → Junction.request
unknown action → keep; switch with green < 8 s → keep

Actuated 9.84 s; reliable mock 10.73 s; 20% bad + shield 11.58 s; 165 bad of 735 calls.

1. Share of bad answers = 165 ÷ 735 = **0.224**, about 22% (set to 20%, drawn at random).

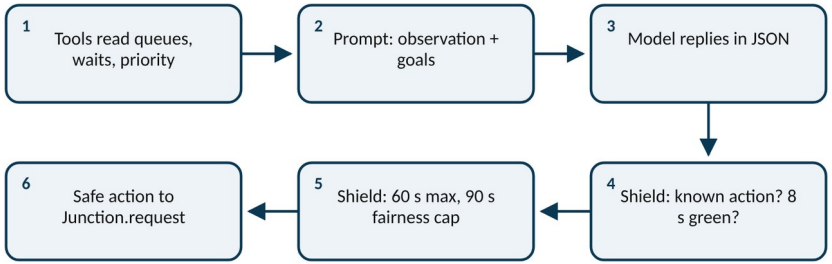
2. Reliable mock vs actuated: 10.73 − 9.84 = **0.89 s** more (+9%).

3. Bad answers + shield vs actuated: 11.58 − 9.84 = **1.74 s** more (+18%).

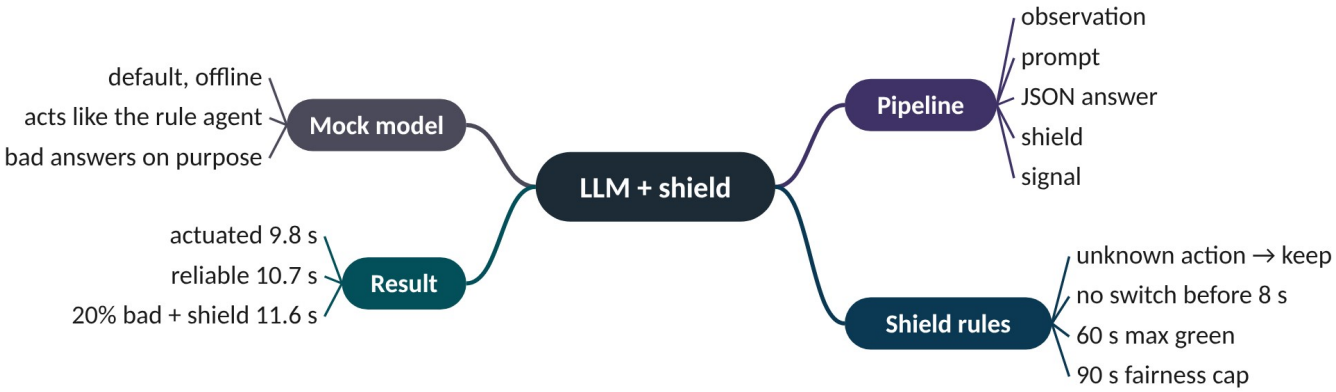
4. Bad answers cost a further 11.58 − 10.73 = **0.85 s**.

5. Unsafe orders reaching the signal: **0**.

One bad answer in five: still safe, and under a second of extra delay.



Remember it: LLM + shield



PICTURE THIS · STOP 7: PARROT CAGE

A talkative parrot shouts orders, some of them nonsense ("green to the north-east!"); a guard reads every order against the rulebook before passing it on.

TRAP

The shield does not make a bad model good: legal but poor answers pass and cost delay.

The controllers you build in Go 2

	Actuated	Rule agent	Q-learning	LLM + shield
Decides with	gaps between vehicles	goal weights and hysteresis	a learned table of keep/switch values	a model's JSON answer
How often	every second	every 2 s	every 5 s	every 5 s
Explains itself?	no	yes, a reason per decision	no	yes, the model's reason
Safety	min / max green	shield	min green only	shield
One-junction delay	9.8 s	9.6 s	11.7 s	10.7 s (11.6 s with 20% bad)

Things beginners mix up

This...	...is not this	The fact that tells them apart
Hysteresis	**Minimum green**	Hysteresis (2 points) is a judgement margin; minimum green (8 s) is a hard rule the shield enforces.
Green wave	**Platoon messages**	A green wave fixes offsets in advance; messages tell the neighbour live what is coming.
SPaT	**GLOSA**	SPaT is what the signal broadcasts; GLOSA is the speed the car works out from it.
Bad answer	**Unsafe action**	The shield blocks rule-breaking answers; legal but poor answers pass and cost delay.
Mock LLM	**Real LLM**	Lab 6 numbers come from a mock that imitates failures; a real model needs the anthropic backend option.

Check yourself (1 of 2)

Why does every controller change the lights only through Junction.request?

A. It is faster than TraCI

B. So all share the same safety timings and comparisons are fair

C. Because SUMO allows only one call per second

D. To save CO2

Q = −4, reward −2, best next Q = −5, α = 0.2, γ = 0.9. What is the new Q?

A. −4.5

B. −5.0

C. −6.5

D. −3.5

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

The model answers {"action": "green_NE"}. What does the shield do?

A. Turns NE green

B. Holds all-red

C. Replaces it with keep

D. Stops the simulation

In Lab 6 the reliable mock (no bad answers) still had 97 answers blocked. Why?

A. The shield is faulty

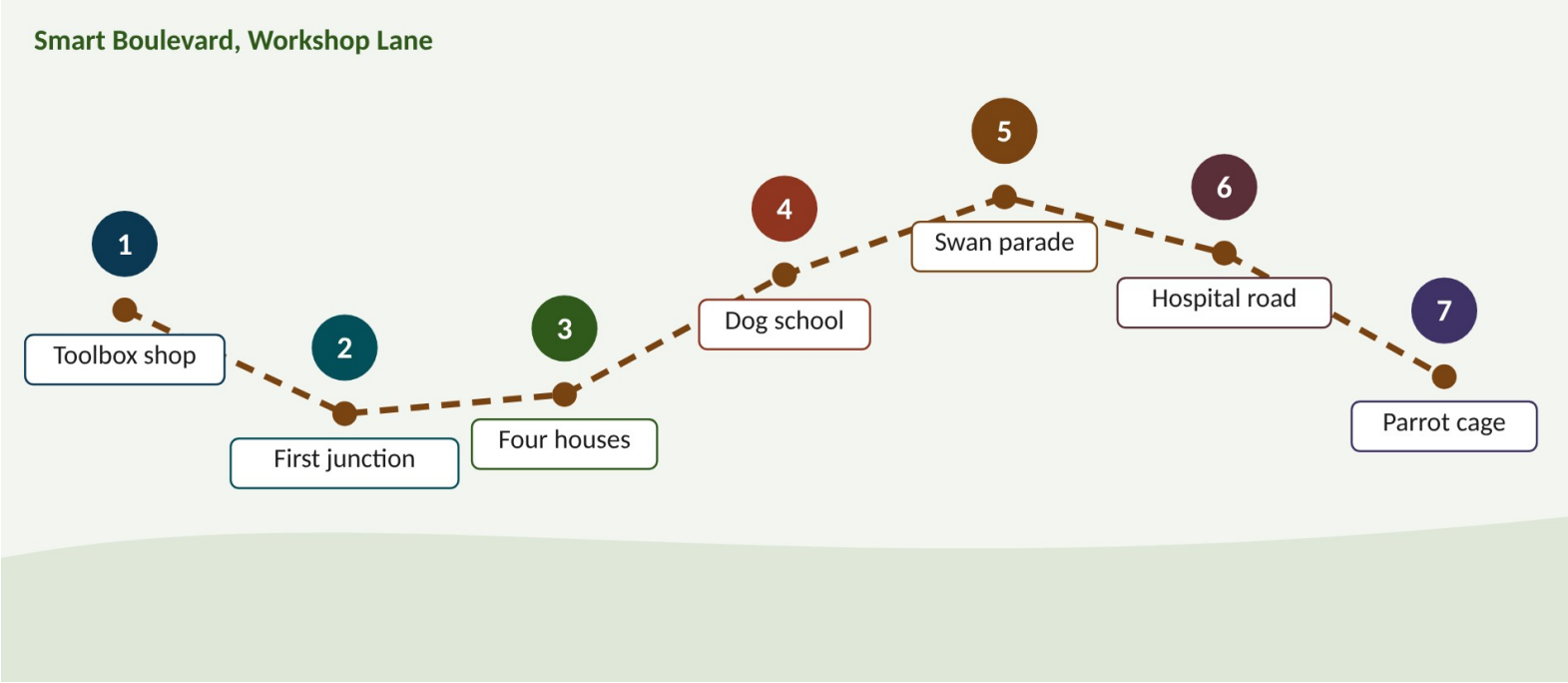
B. The rule reasoning can propose a switch before 8 s of green; the shield enforces it

C. The model invented stages

D. The fairness cap was set to 10 s

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Toolbox shop = Set-up & Junction · 2 First junction = First agent · 3 Four houses = Many agents · 4 Dog school = Q-learning · 5 Swan parade = Eco-driving · 6 Hospital road = Priority tools · 7 Parrot cage = LLM + shield

Numbers to remember

› Yellow 3 s, all-red 1 s, min green 8 s, max green 60 s

› Score = 1·queue + 0.7·arriving + 0.15·wait; switch if ahead by 2

› Lab 1 delay: 14.1 / 9.8 / 15.0 / 9.6 s; longest wait 301 s (MP) vs 20 s

› Lab 2: stops 0.95 → 0.88 with talking; actuated 17.5 s lowest

› Lab 3: α 0.2, γ 0.9; 31 states; 11.7 s vs 9.6 s rules

› Lab 4: v = d ÷ (remaining + 4 s); fixed stops 1.12 → 0.50

› Lab 5: w_priority 25; bus 33.4 → 7.8 s. Lab 6: 165 bad answers, 11.6 s

Keep going

Code, labs and interactive website: smrahmanbd.github.io/agentic-atsc-lab